

Roll No: 2003055

SET - A

Lab Final

Lab Task Q[1]

Question:

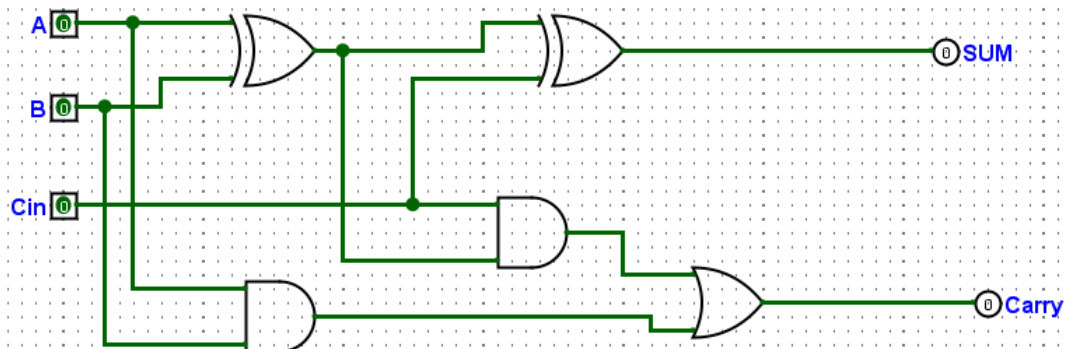
4 bit ALU

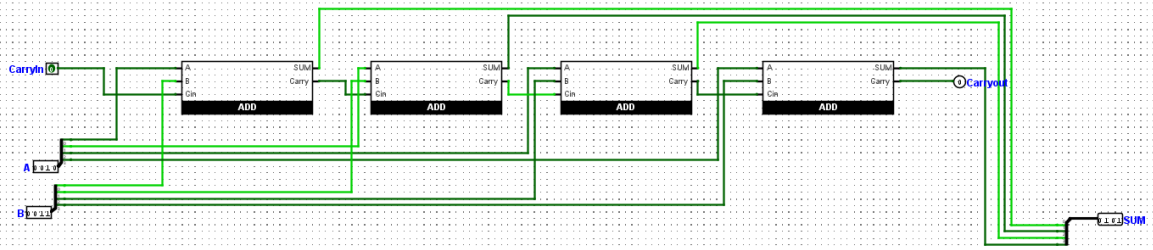
Solution (Code/Add all circuits diagrams):

ORR:

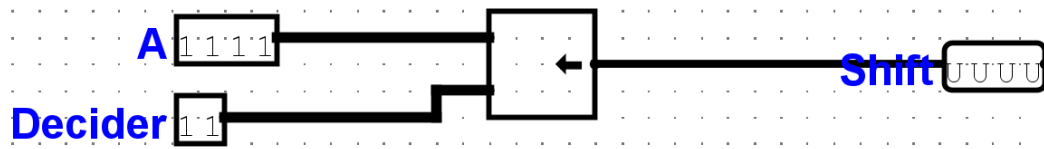


ADD:

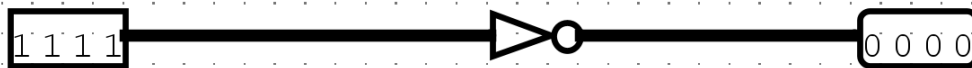




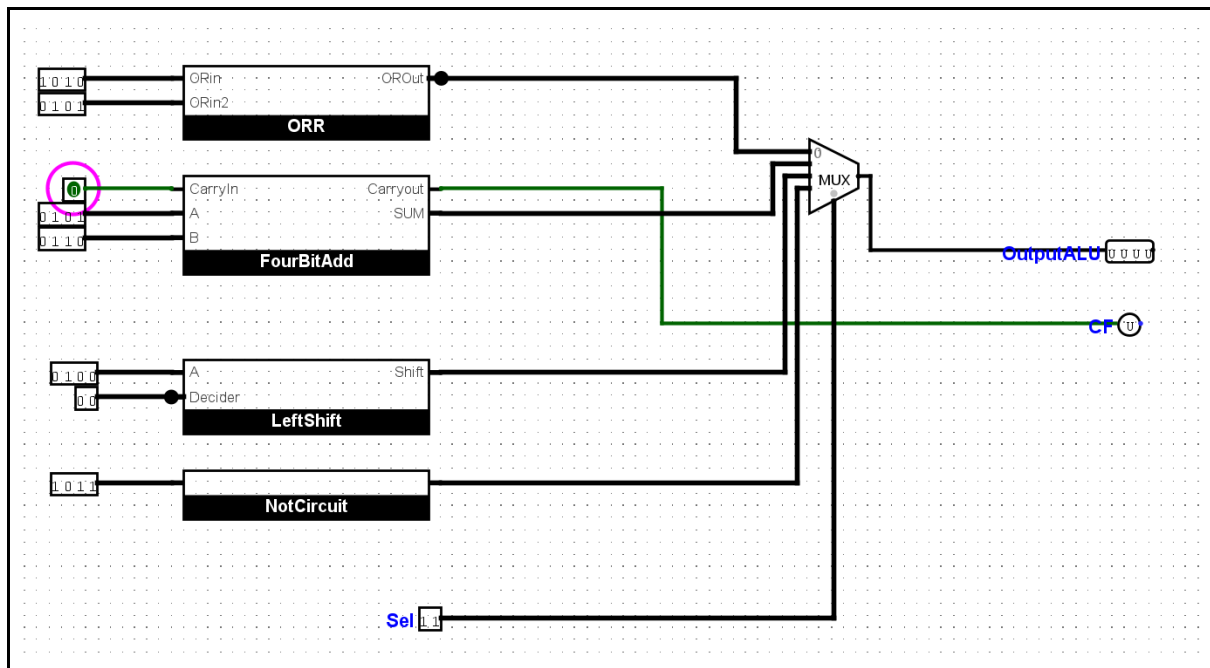
LEFT SHIFT:



NOT:



ALU:



Output/Simulation Result (Screenshot/SnapShot):

Lab Task Q[2]

Question:

Left rotate operation on a 4 bit operand

Solution (Code/Add all circuits diagrams):

```
module left_rotate_4bit (  
    input [3:0] A,          // 4-bit input operand  
    output [3:0] Y          // 4-bit rotated output  
);  
  
    assign Y = {A[2:0], A[3]}; // Left rotate operation  
  
endmodule
```

testbench:

```
`timescale 1ns/1ns  
  
module tb_left_rotate_4bit;  
  
    reg [3:0] A;          // 4-bit input operand  
    wire [3:0] Y;         // 4-bit rotated output  
  
    left_rotate_4bit uut (  
        .A(A),  
        .Y(Y)  
    );  
  
    initial begin  
        $dumpfile("test.vcd");  
        $dumpvars(0, tb_left_rotate_4bit);  
  
        A = 4'b1001; #10; // Test with input 1001  
        A = 4'b1100; #10; // Test with input 1100  
        A = 4'b0111; #10; // Test with input 0111  
        A = 4'b0001; #10; // Test with input 0001  
        $finish;  
    end  
  
    initial begin
```

```
    $monitor("Time = %0t | A = %b | Y = %b", $time, A, Y);  
end  
  
endmodule
```

Output/Simulation Result (Screenshot/SnapShot):

Terminal:

```
PS C:\Users\Asus\Desktop\2003055_Lab_Final> vvp a.out  
VCD info: dumpfile test.vcd opened for output.  
Time = 0 | A = 1001 | Y = 0011  
Time = 10 | A = 1100 | Y = 1001  
Time = 20 | A = 0111 | Y = 1110  
Time = 30 | A = 0001 | Y = 0010
```

gtkWave:

